Lecture 2

Combining Observations

August 27, 2026

Recap from before

Statistics deals with questions of

measurement (how much? how to do it? when to do it?)

evaluation (what do I do with my measurement?)

uncertainty quantification (how do I know that I measured and reported correctly?)

Learning objectives

- Explain why overdetermined problems in astronomy and geodesy created a need for principled ways to combine observations, including averaging, weighting, and least squares.
- Reconstruct and critique Mayer's strategy of grouping and summing 27 equations into three equations, including his comparison with a three-equation solution and his uncertainty calculation.
- Contrast Mayer's relatively comparable observations with Euler's heterogeneous historical data to explain why error structure and model adequacy matter when observations are combined.

1600s: The Quest to Measure the Earth

1. The Big Question

What is the shape and size of the Earth?

2. Better Instruments

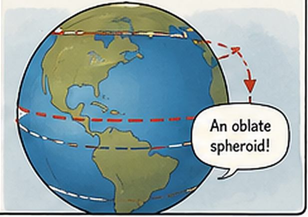
Astronomers like Hevelius and Picard used improved instruments to measure angles in the sky.

3. Measuring Arcs

They measured the length of a degree of latitude by surveying across land.

4. A Surprising Result

They found Earth is not a perfect sphere—it is slightly flattened at the poles.



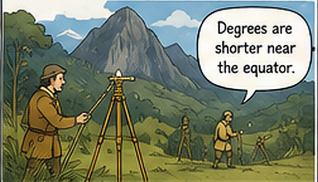
1700s: Great Expeditions, Global Knowledge

1. The French Expedition

In the 1730s, French scientists measured near the Arctic (Maupertuis and team).

2. The Peruvian Expedition

At the same time, others measured near the Equator (La Condamine and team).



3. Solving the Debate

Combining results showed Earth is flattened at the poles. Newton was right!

4. Building Reference Systems

These efforts created the first geodetic frameworks—networks of measured points.

TODAY: GEODESY IN ACTION

1. Satellites Observe Earth

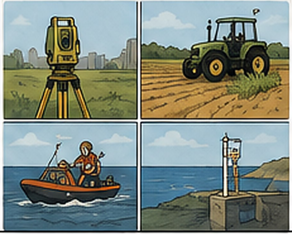
GNSS satellites (GPS, Galileo, GLONASS, BeiDou) orbit Earth and send precise signals.

2. Receivers Measure Position

Your phone or GPS receiver measures signals to determine position, velocity, and time.

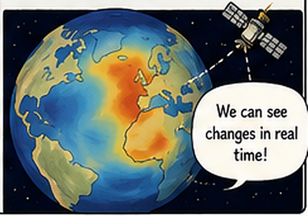
3. Many Uses

From navigation and surveying, to agriculture, disaster response, and tracking sea level rise.



4. Understanding a Changing Earth

Space geodesy (satellite tracking, laser ranging, GRACE) measures Earth's shape, gravity, and how it changes over time.



How can we combine observations?

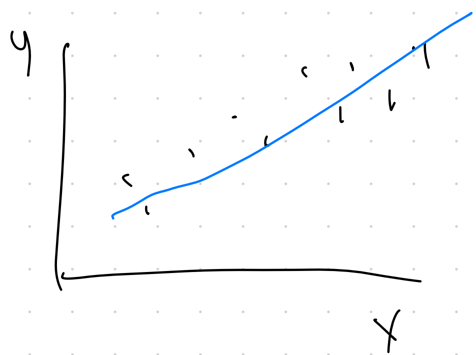
$x_1, x_2, x_3, \ldots$

mean : $(x_1 + x_2 + x_3 + \ldots + x_n)/n$

weighted average : $(w_1 x_1 + \ldots + w_n x_n) / (w_1 + \ldots + w_n)$

"weight"

$(x_1, y_1), \ldots, (x_n, y_n)$



scatter plot

Really quick intro to Legendre's method

"Method of Least Squares"

$$E \ \cancel{0} = a + bx + cy + fz + \dots$$

~~Should be zero~~

"Error"

find $a, b, c, \dots f, \dots$ to minimize E^2

How much data do you even want?

$$\begin{cases} 64 = a + b & (1) \\ 75 = (a+5) + 2(b+1) & (2) \end{cases}$$

$$a = 64 - 4 = 60$$

$$11 = b + 2 + 5 \Rightarrow b = 4$$

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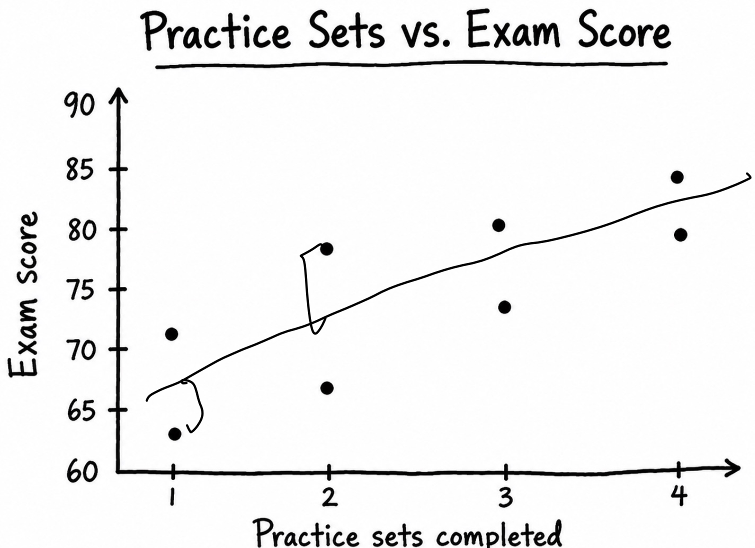
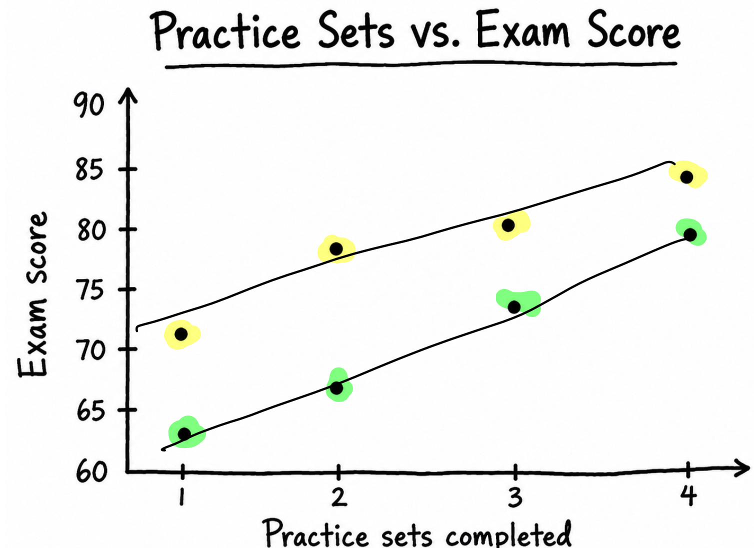
(3) $68 = a + 2b$ ← no new information
still $a = 60$, $b = 4$

(3') $69 = a + 2b$ from linear algebra
no solution for (1, 2, 3')


Practice Sets vs. Exam Score
Exam Score
90
85
80
75
70
65
60
1
2
3
4
Practice sets completed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What's in a mean?

$x_1, \dots, x_n$ data — locations of celestial bodies
$w_1, \dots, w_n$ weights — sizes / mass of the bodies

Cotes — given observations of the same celestial body — where is it really?

$$\frac{x_1 + x_2 + x_3}{3} \qquad \frac{w_1 x_1 + w_2 x_2 + w_3 x_3}{w_1 + w_2 + w_3}$$

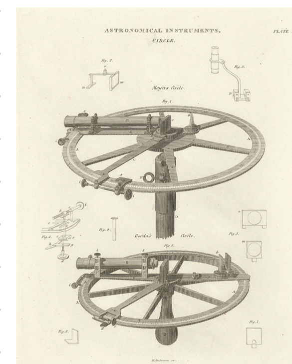
better because it took into account some gravitational info

Tobias Mayer

Life: 1723-1762 —born in Marbach and grew up in very poor conditions.

First book in 1741: Neue und allgemeine Art, alle Aufgaben aus der Geometrie vermittelst der geometrischen Linien leichter insbesondere wie alle reguläre und irreguläre Vielecke, davon ein Verhältnis ihrer Seiten gegeben, in den Circul geometrisch sollen eingeschrieben werden, sammt einer kurzen nötigen Buchstaben-Rechnenkunst und Geometrie.

Spent much of his time on cartography, producing a map of the moon, and relating the measurement of the moon to positions on earth.



Inventor(!) of an improved "reflecting circle" — a navigation tool.

Put together Lunar Tables to help with navigation…

Drama

Mayer's tables submitted for the Longitude Act Award of 10,000 GBP.

Pro Mayer Camp: James Bradley, Astronomer Royal

Competition: John Harrison, watch maker

Why Drama?: Bradley is essentially a judge for the prize

→ Find out what Greenwich time is

→ Find out what current solar time is

we rotate 15° every hour

12 pm

5 pm

75° East of Greenwich

Mayer's geodesy task

How do you measure longitude at sea?

Measure where meridian is

Convert that into knowledge about angles between

Earth + Moon

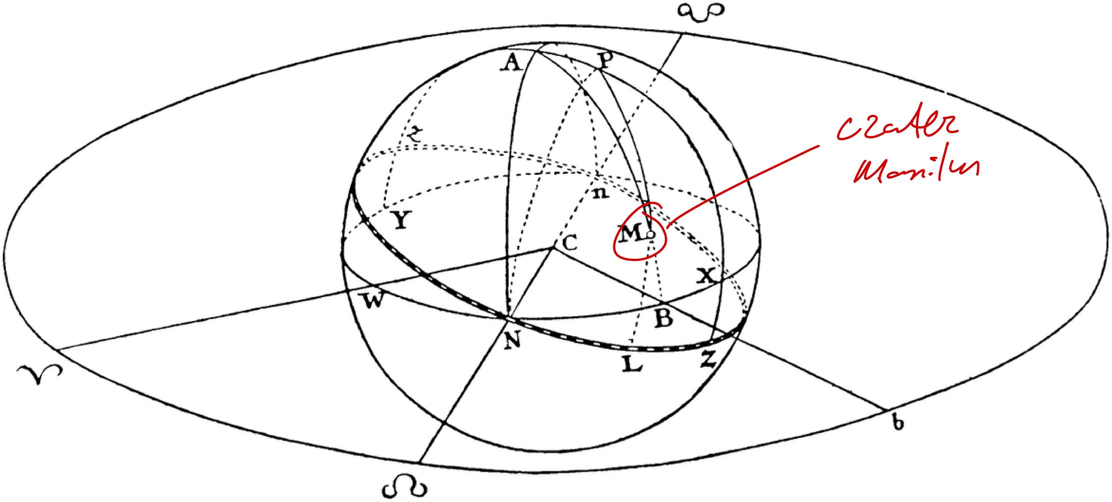


Figure 1.1. Tobias Mayer's original drawing of the moon. (From Mayer, 1750, table VI.)

Mayer's math

$$E = (90 - h) - \beta - \alpha \sin(g - k) - \alpha \sin(\theta) \cos(g - k)$$

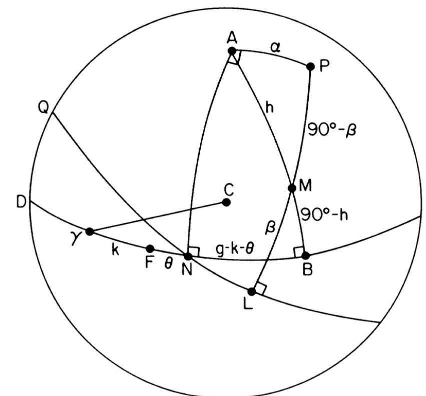


Figure 1.2. The moon. M, The crater Manilius; NL, the moon's equator; P, the moon's equatorial pole; NB, the circumference parallel to the plane of the ecliptic; A, the pole of NB; Cγ, the direction of the equinox from the moon's center C; F, the node of the moon's orbit and the plane of the ecliptic. See text for more details.

issues

- Assumptions about the shape of the moon
  modern "model misspecification"
- imperfect measurements
- External interference with observations

Table 1.1. Mayer's twenty-seven equations of condition, derived from observations of the crater Manilius from 11 April 1748 through 4 March 1749.

Eq. no.	Equation	Group
1	$\beta - 13°10' = +0.8836\alpha - 0.4682\alpha \sin\theta$	
2	$\beta - 13°8' = +0.9996\alpha - 0.0282\alpha \sin\theta$	
3	$\beta - 13°12' = +0.9899\alpha + 0.1421\alpha \sin\theta$	
4	$\beta - 14°15' = +0.2221\alpha + 0.9750\alpha \sin\theta$	
5	$\beta - 14°42' = +0.0006\alpha + 1.0000\alpha \sin\theta$	
6	$\beta - 13°1' = +0.9308\alpha - 0.3654\alpha \sin\theta$	
7	$\beta - 14°31' = +0.0602\alpha + 0.9982\alpha \sin\theta$	
8	$\beta - 14°57' = -0.1570\alpha + 0.9876\alpha \sin\theta$	
9	$\beta - 13°5' = +0.9097\alpha - 0.4152\alpha \sin\theta$	
10	$\beta - 13°2' = +1.0000\alpha + 0.0055\alpha \sin\theta$	
11	$\beta - 13°12' = +0.9689\alpha + 0.2476\alpha \sin\theta$	
12	$\beta - 13°11' = +0.8878\alpha + 0.4602\alpha \sin\theta$	
13	$\beta - 13°34' = +0.7549\alpha + 0.6558\alpha \sin\theta$	
14	$\beta - 13°53' = +0.5755\alpha + 0.8178\alpha \sin\theta$	
15	$\beta - 13°58' = +0.3608\alpha + 0.9326\alpha \sin\theta$	
16	$\beta - 14°14' = +0.1302\alpha + 0.9915\alpha \sin\theta$	
17	$\beta - 14°56' = -0.1068\alpha + 0.9943\alpha \sin\theta$	
18	$\beta - 14°47' = -0.3363\alpha + 0.9418\alpha \sin\theta$	
19	$\beta - 15°56' = -0.8560\alpha + 0.5170\alpha \sin\theta$	
20	$\beta - 13°29' = +0.8002\alpha + 0.5997\alpha \sin\theta$	
21	$\beta - 15°55' = -0.9952\alpha - 0.0982\alpha \sin\theta$	
22	$\beta - 15°39' = -0.8409\alpha + 0.5412\alpha \sin\theta$	
23	$\beta - 16°9' = -0.9429\alpha + 0.3330\alpha \sin\theta$	
24	$\beta - 16°22' = -0.9768\alpha + 0.2141\alpha \sin\theta$	
25	$\beta - 15°38' = -0.6262\alpha - 0.7797\alpha \sin\theta$	
26	$\beta - 14°54' = -0.4091\alpha - 0.9125\alpha \sin\theta$	
27	$\beta - 13°7' = +0.9284\alpha - 0.3716\alpha \sin\theta$	

Source: Mayer (1750, p. 153).
Note: One misprinted sign in equation 7 has been corrected.

Mayer's approach

Group the equations

Sum up equations within a group

Solve 3 eqn's in 3 unknowns

Estimates α $1^\circ 30'$

Groups are "similar" because he is the instrument

Extra Slide – Mayer's method

$$0 = a x_{11} + b x_{21}$$
$$0 = a x_{12} + b x_{22}$$
$$0 = a x_{13} + b x_{23}$$

$$1 = a y_{11} + b y_{21}$$
$$1 = a y_{12} + b y_{22}$$
$$1 = a y_{13} + b y_{23}$$

Add then up

$$0 = a\underbrace{(x_{11}+x_{12}+x_{13})} + b\underbrace{(x_{21}+x_{22}+x_{23})}$$
$$= a \cdot \tilde{x}_1 + b \cdot \tilde{x}_2$$

$$3 = a\underbrace{(y_{11}+y_{12}+y_{13})} + b\underbrace{(y_{21}+y_{22}+y_{23})}$$
$$= a \cdot \tilde{y}_1 + b \cdot \tilde{y}_2$$

Rearrange terms

$$-b\tilde{x}_2 = a\tilde{x}_1 \qquad 3 - b\tilde{y}_2 = a\tilde{y}_1$$

Divide through

$$-b\tilde{x}_2/\tilde{x}_1 = a\tilde{x}_1/\tilde{x}_1 = a \qquad (3 - b\tilde{y}_2)/\tilde{y}_1 = a\tilde{y}_1/\tilde{y}_1 = a$$

Combine equations
Solve for b

$$-b\tilde{x}_2/\tilde{x}_1 = (3 - b\tilde{y}_2)/\tilde{y}_1$$
$$(-b\tilde{x}_2/\tilde{x}_1)\tilde{y}_1 = 3 - b\tilde{y}_2$$
$$b\tilde{y}_2 - b(\tilde{x}_2\tilde{y}_1/\tilde{x}_1) = 3$$
$$b(\tilde{y}_2 - \tilde{x}_2\tilde{y}_1/\tilde{x}_1) = 3$$
$$b = \frac{3}{(\tilde{y}_2 - \tilde{x}_2\tilde{y}_1/\tilde{x}_1)}$$

Plug in for solving for a:

$$a = \frac{-\tilde{x}_2}{\tilde{x}_1}\left(\frac{3}{\tilde{y}_2 - \tilde{x}_2\tilde{y}_1/\tilde{x}_1}\right)$$

Mayer's accuracy determination

His method produced one estimate of α

He had a "naive" estimate of α as well

Claim: his method is more accurate so use contrast to quantify this.

Mayer's estimate	Naive estimate
$1^\circ 30'$	$1^\circ 40'$

uncertainty: $1^\circ 30' \pm X$

Contrast $\quad 1^\circ 40' - (1^\circ 30' \pm X) = 10' \pm X$

Claim $\quad (\pm X) \times 9 = (10' \pm X)$

why 9?

because he used 9 times the data

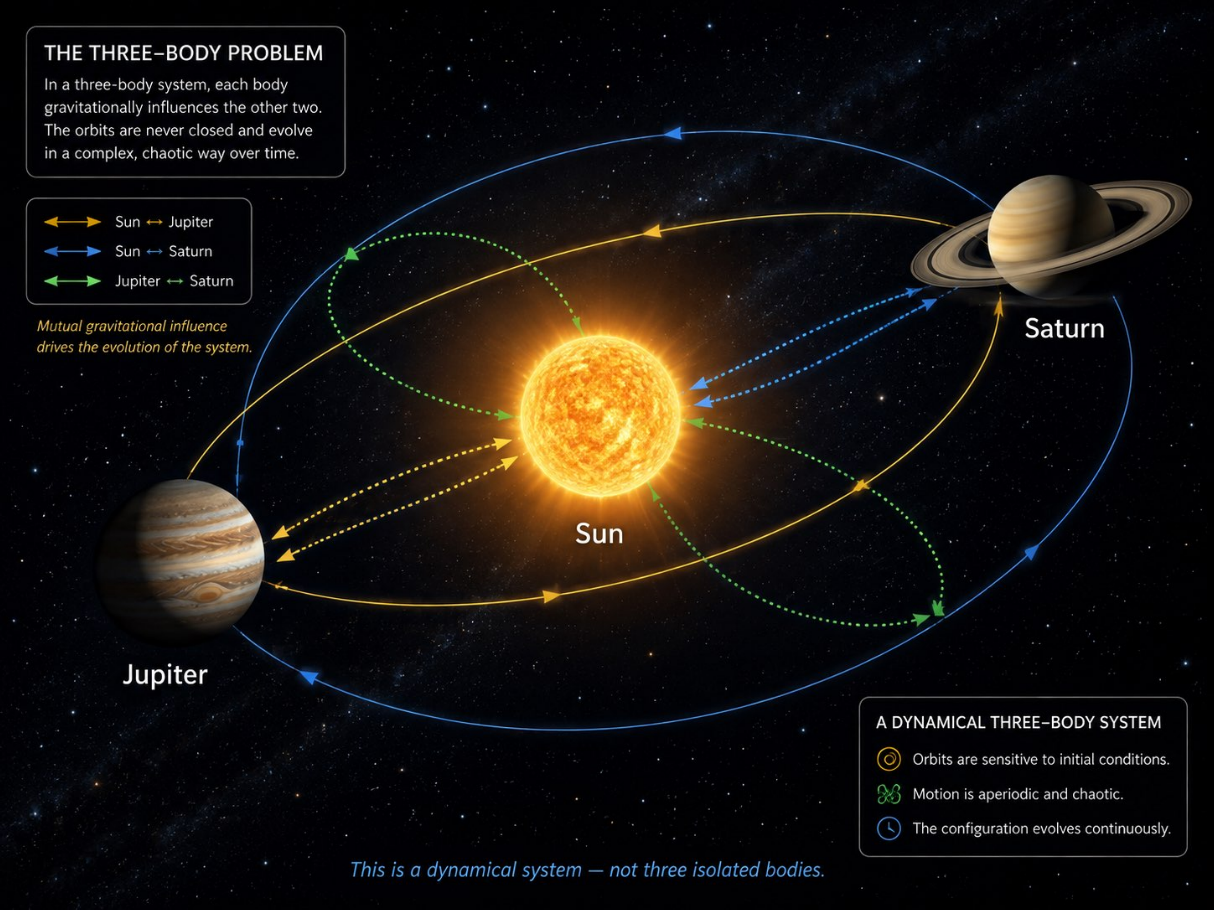
(recall that this is actually wrong ...)

Euler's failures

9 unknowns in his equations
Lots of heterogeneity in observation quality.

Unable / unwilling to combine observations

Could "multiply errors"



THE THREE–BODY PROBLEM
In a three-body system, each body gravitationally influences the other two. The orbits are never closed and evolve in a complex, chaotic way over time.
Sun ↔ Jupiter
Sun ↔ Saturn
Jupiter ↔ Saturn
Mutual gravitational influence drives the evolution of the system.
Saturn
Sun
Jupiter
A DYNAMICAL THREE–BODY SYSTEM
Orbits are sensitive to initial conditions.
Motion is aperiodic and chaotic.
The configuration evolves continuously.
This is a dynamical system — not three isolated bodies.

What are different errors?

<u>errors</u> <u>random errors</u>

... we will come back to this.

What *would* class have looked like with a student led discussion?

Discussion ~20 mins

1. Mayer biography
2. What does it mean for observations to be equivalent? (in general, specifically for Mayer)
3. Why did Euler not have the same "equivalence"?
4. What are "errors" vs "random errors"?

Rest of class

5. Historical context for geodesy
6. Intro to some notation
7. Mayer's method
8. Euler's "method"

todays lecture: 9-31, total covered: 1-31

For NEXT CLASS

Read: 32-54

Think about: What makes a good generalizable principle